[Project #1] Personal Portfolio

Feedback Deadline: Apr 23, 2025 11:59 PM CST
Cutoff Date: Apr 30, 2025 11:59 PM CST(No feedback will be given)
GitHub Classroom: https://classroom.github.com/a/eAJQqnMG (Please don't remove the feedback PR)
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Objective: Create a personal portfolio using Next.js.

Overview

This is an opportunity for you to create a web app that can showcase your creativity and frontend skills.

Requirements

1. **Build your own components instead of using UI libraries like Material UI**
2. Your portfolio should not be static. It should incorporate **several** opportunities for user interaction (ex: form, toggling light/dark theme, comments, be creative!).
3. Must be more than 1 page - create several pages for various topics that you want to showcase.
4. [Stretch] Store user preferences – this is related to #1. Ex: If I toggle the light/dark theme, my preference should stay with me throughout the app.
    a. [Bonus] - The preference is persistent even after refreshing the page.
5. Must be accessible – use proper semantics and tooling to ensure that your portfolio is usable by all.
6. Mini Design Doc
    a. General template:
        i. Overview/Background
        ii. **Technical** problem statement
        iii. Design Approach
            1. Walk me through your thought process:
                a. Why did you choose to separate out a specific section into a component?
                b. Why did you keep a state within a component vs using dependency injection?
                c. Which design patterns did you notice in your app?
                    i. Ex: By default, the module design pattern is used for components

d. What were several challenges that you encountered and how did you solve said challenges?
e. Did you use CSR or SSR or hybrid and why?
f. Anything else? Ex: Using AI tooling (ChatGPT, Bard, etc)

For Self-Evaluation

1. Code quality
2. Components
    a. **Build your own components instead of using libraries like Material UI**
    b. Appropriately breaking your code up into components (https://atomicdesign.bradfrost.com/chapter-2/)
    c. See starter code
3. State handling
4. [Stretch] Dependency Injection (using the context API)
    a. **Should be built from scratch (ex: theming built with 3P tools doesn't count)**
    b. For more info on contexts, see https://react.dev/learn/passing-data-deeply-with-context
5. Event handling
6. Semantics
7. An understanding of why your code works (ex: rendering, the DOM, etc)
    a. You can showcase this in your mini design doc

Deliverable (place all deliverables in your GitHub classroom repo)

1. Your code – no need to build, just include the files for your app.
    a. Be sure to add a .gitignore file to exclude the node_modules directory
2. Link to your mini design doc in the README – be sure that your teaching team has viewer access.

FAQs

What if I already have a portfolio app?

8. That's awesome! Please let me know in the design doc. Some additional challenges for you:

a. Build the portfolio using a different framework
b. Did you follow best practices when it comes to rendering?
c. Is it polished enough that I can get a signal that you have a solid understanding of software engineering principles (ex: separation of concerns, clean code, etc)?
d. Add a fun/new feature
e. Is it built with accessibility in mind?

Appendix

Next.js Local Dev Setup

1. Install VSCode (https://code.visualstudio.com/download

Links to an external site.

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2. Follow the "Automatic Installation" steps from the Next.js Getting Started docs.

Links to an external site.

- Make sure your system requirements are met (ex: install Node/npm if needed)
- Answer the prompts as the following:
    - What is your project named? my-app
      Would you like to use TypeScript? No
      Would you like to use ESLint? Yes
      Would you like to use Tailwind CSS? Yes
      Would you like to use `src/` directory? Yes
      Would you like to use App Router? (recommended) Yes
      Would you like to customize the default import alias (@/*)? Yes
      What import alias would you like configured? @/*

3. Be sure that you can run your local dev server (Run the Development Server).

Links to an external site.

4. Double check that ESLint is set up. This ensures that your code is formatted properly and checks for potential errors -- we call this **linting.**

- https://nextjs.org/docs/app/building-your-application/configuring/eslint
- Links to an external site.

    - Note: The linked docs have a lot of info -- this is an exercise to get you used to sifting through information just to find what you need.

5. Make sure your VSCode is configured to auto-format whenever you save a file

- To prepare you for using AI tools effectively, try using ChatGPT or Gemini for this one (example prompt: "how do I set up vscode to auto format on save?"). Your response should mention Prettier.
    - Sometimes AI tools will give you the right tool with the wrong steps. When in doubt, double check with the actual docs (Prettier docs in this case).